💻

SWE

Waterfall, XP

1. What is a software development *methodology*?

Answer:

A software development methodology is a formal process or set of practices used to develop software in a systematic and organized way.

---

2. What is Waterfall Model?

Answer:

The Waterfall Model is a traditional software development process where each phase is completed before moving to the next phase.

3. What is Agile Development?

Answer:

Agile is a software development methodology that focuses on continuous delivery, customer collaboration, and adapting to changes.

- Extreme Programming (XP)
- Scrum
- Crystal
- Feature Driven Development (FDD)

PART 1 (XP Principles 1–6)

1. Metaphor

Definition (Exam)

Metaphor is an XP practice that uses a simple, shared story or analogy to help all team members understand how the software system works and communicate using the same concepts.

Explanation

In software development, different developers may understand the system differently, which can create confusion. Metaphor solves this problem by providing a common story or model that everyone follows. It creates a shared language among developers and customers and helps the team understand where new features fit into the system.

Advantages

- Everyone understands the system in the same way.
- Improves communication among team members.
- Reduces misunderstandings.
- Helps new developers learn the project quickly.
- Makes system design more consistent.

Disadvantages

- Finding a good metaphor is difficult.
- A wrong metaphor may confuse the team.
- It may not work well for very large or complex systems.

Example

Suppose a team is developing an online banking system.

They use **Bank** as the metaphor.

- Customer → Account Holder

- Money → Balance
- Transfer → Transaction

Since everyone uses the same concept, communication becomes easier.

---

5 Marks Answer

Metaphor is one of the key practices of Extreme Programming (XP). It uses a simple, shared story or analogy to explain how the software system works. It provides a common language for both developers and customers, making communication easier and reducing confusion. It also helps new developers understand the system quickly and ensures consistency throughout the project. However, choosing a suitable metaphor can be difficult, and an inappropriate metaphor may create misunderstandings. A banking application using the concept of a real bank is a good example of metaphor.

---

2. Release Planning (Planning Game)

Definition

Release Planning is an XP practice in which customers and developers work together to decide which features will be developed first and when they will be released.

Explanation

Customers write their requirements as ***User Stories.***

Developers estimate the time, effort, and risk for each story.

Then, both decide which features should be included in the next release.

Advantages

- Customers get high-priority features first.
- Better planning.

- Reduces project risk.
- Improves customer satisfaction.
- Makes project management easier.

Disadvantages

- Wrong estimates can delay the project.
- Customer participation is always required.
- Frequent planning meetings take time.

Example

Customer wants

- Login
- Payment
- Chat

Developers estimate

Login = 2 days

Payment = 7 days

Chat = 10 days

So Login is released first.

---

5 Marks Answer

Release Planning, also called the Planning Game, is an XP practice where customers and developers work together to decide the order of software features. Customers provide user stories, and developers estimate the effort and risk required for implementation. The team then selects the highest-priority features for the next release. This practice improves planning, reduces project risk, and increases customer satisfaction. However, inaccurate estimation and continuous customer involvement may become challenges.

---

3. Testing (Test-Driven Development)

Definition

Test-Driven Development (TDD) is an XP practice where *developers write test cases before* writing the actual program code.

Explanation

Normally, developers write code first.

In XP:

First → Test

Then → Code

The code is accepted only if all tests pass.

Advantages

- Produces fewer bugs.
- Improves software quality.
- Detects errors early.
- Makes future maintenance easier.
- Improves code design.

Disadvantages

- Takes more time initially.
- Writing tests requires experience.
- Not suitable for every project.

Example

Before writing a calculator program,

the developer first writes a test:

2 + 2 = 4

Then writes the code until the test passes.

---

5 Marks Answer

Test-Driven Development (TDD) is an important XP practice where developers write automated test cases before writing the program code. After writing the code, all tests must pass before moving to the next task. TDD helps detect errors early, improves software quality, and makes maintenance easier. However, it increases development time initially and requires skilled developers. A calculator application where test cases are written before coding is a common example.

---

4. Pair Programming

Definition

Pair Programming is an XP practice where two programmers work together on one computer to develop software.

Explanation

One programmer is called the ***Driver.***

They write the code.

The other programmer is the ***Navigator.***

They review the code, find mistakes, and suggest improvements.

They regularly exchange roles.

Advantages

- Produces better-quality software.
- Finds bugs quickly.

- Improves teamwork.
- Increases knowledge sharing.
- Reduces coding mistakes.

Disadvantages

- Higher development cost.
- Not effective if partners do not cooperate.
- May reduce productivity for simple tasks.

Example

Rahim writes code.

Karim checks every line and immediately points out mistakes.

After one hour, they switch roles.

5 Marks Answer

Pair Programming is a software development practice in XP where two programmers work together using one computer. One programmer, called the Driver, writes the code, while the other, called the Navigator, reviews the code and provides suggestions. The two programmers regularly switch roles. Pair Programming improves software quality, reduces bugs, and encourages teamwork and knowledge sharing. However, it requires more resources and may not be suitable for simple programming tasks.

5. Refactoring

Definition

Refactoring is the process of improving the internal structure of existing code without changing its external behavior or functionality.

Explanation

Sometimes software works correctly,

but the code is difficult to understand.

Refactoring improves the code structure,

removes duplicate code,

and makes maintenance easier,

while keeping the output the same.

Advantages

- Cleaner code.
- Easier maintenance.
- Improves readability.
- Removes duplicate code.
- Better software design.

Disadvantages

- Takes additional development time.
- Incorrect refactoring may introduce new bugs.
- Requires experienced developers.

Example

A function contains 500 lines.

After refactoring,

it becomes 10 small functions.

The output remains exactly the same.

---

5 Marks Answer

Refactoring is an XP practice that improves the internal structure of software without changing its functionality. It removes duplicate code, improves readability, and simplifies maintenance. Refactoring helps developers produce cleaner and more organized software. However, it requires additional time and experienced developers, and incorrect refactoring may introduce new defects. Dividing a large function into several smaller functions is a common example of refactoring.

---

6. Simple Design

Definition

Simple Design is an XP practice that focuses on designing only the features that are currently needed while avoiding unnecessary complexity.

Explanation

XP believes developers should not design for future requirements that may never be needed.

Instead,

they should create the simplest design that works today.

Future improvements can be added later through refactoring.

Advantages

- Faster development.
- Lower cost.
- Easier maintenance.
- Less unnecessary work.
- Faster customer feedback.

Disadvantages

- Future expansion may require redesign.
- Poor planning may affect scalability.
- Not suitable if future requirements are already known.

Example

A company currently needs only Login.

Instead of building Face ID,

Voice Login,

and Fingerprint Login,

developers implement only the Login feature.

---

5 Marks Answer

Simple Design is an XP practice that encourages developers to build only what is necessary for the current release. It avoids unnecessary complexity and helps deliver working software quickly. Future improvements can be made later using refactoring. This practice reduces development cost, speeds up delivery, and simplifies maintenance. However, if future requirements are ignored completely, redesign may become necessary later.

---

7. Collective Code Ownership

Definition

Collective Code Ownership is an XP practice where every developer has the right and responsibility to modify any part of the project's source code whenever necessary.

Explanation

In many projects, a specific developer owns a specific module. If that developer is absent, others cannot fix problems easily.

XP removes this limitation by making the code the property of the whole team. Any developer can improve or fix any part of the code.

Advantages

- Faster bug fixing.
- No dependency on one developer.
- Better teamwork.
- Improves code quality.
- Faster project progress.

Disadvantages

- Developers may accidentally change others' code.
- Poor communication can create conflicts.
- Requires strong coding standards.

Example

Rahim wrote the Login module.

Karim finds a bug.

Instead of waiting for Rahim, Karim fixes the bug immediately.

---

5 Marks Answer

Collective Code Ownership is an XP practice in which the source code belongs to the whole development team rather than an individual programmer. Every developer can modify any part of the project whenever necessary. This practice speeds up development, improves teamwork, and allows bugs to be fixed quickly.

However, without proper communication and coding standards, developers may accidentally introduce new problems into the code.

8. Continuous Integration (CI)

Definition

Continuous Integration (CI) is an XP practice where developers frequently integrate their tested code into the main project repository.

Explanation

Instead of waiting until the end of the project,

developers merge their code several times a day.

Before merging,

all unit tests must pass successfully.

This helps detect integration problems early.

Advantages

- Detects bugs early.
- Reduces merge conflicts.
- Keeps software stable.
- Improves software quality.
- Faster delivery.

Disadvantages

- Requires automated testing.
- Frequent integration needs discipline.
- Initial setup can be expensive.

Example

Three developers finish different features.

Instead of merging after one month,

they merge their code every day.

Problems are found immediately.

---

5 Marks Answer

Continuous Integration is an XP practice where developers frequently merge their tested code into the main project. Every integration is followed by automated testing to ensure the software remains stable. This practice reduces integration problems, detects bugs early, and improves software quality. However, it requires automated testing tools and disciplined development practices.

---

9. On-site Customer

Definition

On-site Customer is an XP practice where the customer works closely with the development team throughout the project to answer questions and make business decisions.

Explanation

Instead of waiting for emails or meetings,

the customer stays available.

Whenever developers have questions,

they receive immediate answers.

This reduces misunderstanding.

Advantages

- Better communication.
- Quick decision-making.
- Correct requirements.
- Higher customer satisfaction.
- Fewer misunderstandings.

Disadvantages

- Customers are often busy.
- Customers may not always know what they want.
- Difficult to keep customers available all the time.

Example

Developer asks,

"What should happen after payment?"

Customer immediately answers,

"Show the payment success page."

Development continues without delay.

5 Marks Answer

On-site Customer is an XP practice in which the customer remains available to the development team throughout the project. The customer clarifies requirements, answers questions, and makes business decisions quickly. This improves communication, reduces misunderstandings, and increases customer satisfaction. However, keeping customers available throughout the project is often difficult because they may have other responsibilities.

10. Small Releases

Definition

Small Releases is an XP practice that delivers software in small and frequent versions instead of one large release.

Explanation

Rather than waiting several months,

developers release a small working version every few weeks.

Customers use it,

give feedback,

and improvements are made in the next release.

Advantages

- Early customer feedback.
- Lower project risk.
- Faster delivery.
- Easier bug detection.
- Higher customer satisfaction.

Disadvantages

- Requires continuous planning.
- Frequent deployment takes effort.
- Customers may expect updates too often.

Example

Instead of releasing 20 features after six months,

developers release five features every month.

5 Marks Answer

Small Releases is an XP practice that delivers software in small and frequent versions. Each release provides useful functionality and allows customers to give early feedback. This practice reduces project risk, improves customer satisfaction, and helps developers detect problems quickly. However, it requires continuous planning and regular deployment activities.

---

11. Forty-Hour Work Week

Definition

The Forty-Hour Work Week is an XP practice that encourages developers to work reasonable hours and avoid excessive overtime.

Explanation

XP believes tired developers make more mistakes.

Working normal hours keeps developers fresh,

productive,

and motivated.

Good planning is better than continuous overtime.

Advantages

- Better productivity.
- Fewer mistakes.
- Better work-life balance.
- Happier developers.
- Improved software quality.

Disadvantages

- Difficult during emergencies.
- Tight deadlines may require overtime.
- Not always practical in large companies.

Example

Instead of working 16 hours every day,

developers work 8 hours,

rest properly,

and produce better-quality software.

5 Marks Answer

The Forty-Hour Work Week is an XP practice that encourages developers to maintain a healthy work-life balance by avoiding excessive overtime. Well-rested developers make fewer mistakes, produce better-quality software, and remain productive for longer periods. Although emergencies may sometimes require overtime, XP recommends proper planning instead of overworking developers.

12. Coding Standards

Definition

Coding Standards are a set of rules and conventions that developers follow to write consistent, readable, and maintainable source code.

Explanation

Every developer follows the same rules for:

- Variable naming
- Code formatting
- Indentation

- Comments
- File organization

This makes the code easier to understand and maintain.

Advantages

- Improves readability.
- Easier maintenance.
- Better teamwork.
- Reduces coding mistakes.
- Makes debugging easier.

Disadvantages

- Developers may feel restricted.
- Learning standards takes time.
- Strict rules may reduce flexibility.

Example

Instead of writing

```
int x;
```

developers write

```
int studentAge;
```

The second name clearly explains the purpose of the variable.

---

5 Marks Answer

Coding Standards are a set of rules that developers follow to write consistent, readable, and maintainable software code. These standards include naming conventions, formatting rules, indentation, and documentation practices. Coding standards improve teamwork, reduce errors, and simplify software maintenance. However, developers must spend time learning the standards, and some may feel restricted by strict rules.

Stand-Up Meeting (The 13th Practice of XP)

Definition

The Stand-Up Meeting is the 13th practice of Extreme Programming (XP). It is a short daily meeting, *usually lasting 15 minutes,* where all team members stand together to discuss their progress, plans, and any obstacles they are facing.

---

5 Marks Exam Answer

The **Stand-Up Meeting** is considered the **13th practice of Extreme Programming (XP)**. It is a short daily meeting that usually lasts **15 minutes**. During the meeting, all team members stand in a circle, which helps keep the meeting brief and focused.

Each team member answers three important questions:

1. **What did I accomplish yesterday?**
2. **What do I plan to accomplish today?**
3. **What obstacles or challenges am I currently facing?**

The purpose of the Stand-Up Meeting is to improve communication, monitor daily progress, identify problems at an early stage, and ensure that everyone is working toward the same goal. It also helps the team coordinate tasks effectively and can be used to determine pair programming partners for the day. As a result, the Stand-Up Meeting increases collaboration, transparency, and overall team productivity.

NEW CHAPTER. (SCRUM)

1. What is Scrum?

Definition (Exam)

Scrum is a lightweight, iterative, and incremental Agile process model used for managing and developing complex software projects. It focuses on teamwork, continious improvement, and frequent delivery of working software.

2. Three Pillars of Scrum

Scrum is based on **three pillars**:

1. Transparency
2. Inspection
3. Adaptation

Transparency

Answer:

Transparency is the first pillar of Scrum. It means that all project processes, progress, and important information are visible and understandable to everyone involved in the project. Scrum artifacts such as the ***Product Backlog and Sprint***

Backlog help maintain transparency. This improves communication, builds trust among team members, and ensures that everyone works toward the same goal.

Inspection

Answer:

Inspection is the second pillar of Scrum. It refers to regularly reviewing the product, project progress, and development process to identify problems or deviations at an early stage. Scrum events such as the ***Daily Scrum and Sprint Review*** are used for inspection. Regular inspection helps improve software quality and keeps the project on the right track.

Adaptation

Answer:

Adaptation is the third pillar of Scrum. It means making necessary changes to the product or development process based on the results of inspection. If problems are identified or customer requirements change, the Scrum Team adjusts its plans or backlog accordingly. Adaptation enables continuous improvement and helps deliver better-quality software that meets customer needs.

#Scrum Roles

Answer:

A **Scrum Team** consists of **three roles**:

1. Product Owner
2. Scrum Master
3. Developers

Each role has specific responsibilities and works together to achieve the Sprint Goal and deliver a valuable product.

---

1. Product Owner

Answer:

The **Product Owner** is responsible for ***maximizing the value of the product*** developed by the Scrum Team. The Product Owner manages and prioritizes the ***Product Backlog***, defines the ***Product Goal***, communicates customer requirements, and ensures that the team is working on the most valuable features. The Product Owner acts as the bridge between stakeholders and the Scrum Team, ensuring that customer needs are properly understood and implemented.

---

2. Scrum Master

Answer:

The **Scrum Master** is responsible for ensuring that ***Scrum principles, values, and practices are properly followed.*** The Scrum Master acts as a facilitator and coach, helping the Scrum Team work effectively by removing obstacles, organizing Scrum events, and supporting both the Developers and the Product Owner. ***The Scrum Master is not a project manager*** but a servant leader who helps the team continuously improve its performance.

---

3. Developers

Answer:

Developers are the members of the Scrum Team who are responsible for ***building and delivering a usable Product Increment during each Sprint.*** They plan their work, estimate tasks, maintain product quality, collaborate with other team members, and ensure that the ***Sprint Goal*** is achieved. Developers are self-managing and work together to create high-quality software.

#Scrum Events

Answer:

Scrum Events are structured meetings that help the Scrum Team plan, inspect, adapt, and improve their work throughout the Sprint. Scrum consists of **five events**:

1. Sprint
2. Sprint Planning
3. Daily Scrum
4. Sprint Review
5. Sprint Retrospective

---

1. Sprint

Answer:

A Sprint is a fixed time-boxed iteration, usually one month or less, during which the Scrum Team develops a usable and valuable ***Product Increment.*** Each Sprint has a clear ***Sprint Goal***, and no changes are made that would endanger this goal. At the end of every Sprint, a working product increment is delivered.

---

2. Sprint Planning

Answer:

Sprint Planning is the first event of every Sprint, where the Scrum Team decides the ***Sprint Goal,*** selects the ***Product Backlog*** items to be completed, and creates a plan for achieving the Sprint Goal. This meeting ensures that all team members understand the work to be completed during the Sprint.

---

3. Daily Scrum

Answer:

The **Daily Scrum** is a short daily meeting, usually lasting ***15 minutes***, where the Developers inspect their progress toward the ***Sprint Goal*** and plan the work for the next 24 hours. Team members discuss completed work, current tasks, and

any obstacles affecting their progress. This meeting improves communication and helps identify problems early.

---

4. Sprint Review

Answer:

The **Sprint Review** is held at the end of each Sprint to present the ***completed Product Increment to stakeholders and collect their feedback.*** The Scrum Team discusses what has been completed, reviews the product, and updates the Product Backlog based on stakeholder suggestions. This event ensures that the product continues to meet customer needs.

---

5. Sprint Retrospective

Answer:

The Sprint Retrospective is the final event of a Sprint, where the Scrum Team reflects on the completed Sprint to identify what went well, what problems occurred, and how the team can improve in the next Sprint. The goal of this meeting is to continuously improve teamwork, processes, and overall performance.

#Scrum Artifacts

Answer:

Scrum Artifacts are important documents or work items that provide transparency and help the Scrum Team manage, track, and deliver the product effectively. Scrum consists of **three artifacts**:

1. Product Backlog
2. Sprint Backlog
3. Increment

Each Scrum Artifact has a corresponding commitment:

- **Product Backlog → Product Goal**
- **Sprint Backlog → Sprint Goal**
- **Increment → Definition of Done (DoD)**

1. Product Backlog

Answer:

The **Product Backlog** is an ordered and prioritized list of all the ***features, requirements, enhancements, and bug fixes needed for a product.*** It is created and managed by the **Product Owner** and is continuously updated throughout the project. The Product Backlog serves as the single source of work for the Scrum Team and helps ensure that the most valuable features are developed first. ***Its commitment is the Product Goal.***

---

2. Product Goal

Answer:

The **Product Goal** is the long-term objective that the Scrum Team aims to achieve through the development of the product. ***It provides a clear direction for the Product Backlog and ensures that all development activities contribute to the overall vision of the product.*** The Product Goal helps the Scrum Team remain focused on delivering maximum value to customers.

---

3. Sprint Backlog

Answer:

The Sprint Backlog is the set of Product Backlog items selected for a specific Sprint, along with a plan for delivering them. It is ***created and managed by the Developers*** during Sprint Planning. The Sprint Backlog helps the team monitor daily progress and stay focused on achieving the Sprint Goal. Its commitment is the **Sprint Goal**.

---

4. Sprint Goal

Answer:

The **Sprint Goal** is the objective that the Scrum Team plans to achieve during a Sprint. It provides direction and purpose for the Sprint and helps Developers understand why they are building the selected Product Backlog items. The Sprint Goal keeps the team focused on delivering a valuable Product Increment by the end of the Sprint.

---

5. Increment

Answer:

An **Increment** is the completed and usable version of the product created at the end of a Sprint. It consists of all completed Product Backlog items that meet the **Definition of Done (DoD)**. Each Increment adds new functionality to the existing product and must be in a releasable state, allowing it to be delivered to customers if required. *Its commitment is the **Definition of Done**.*

---

6. Definition of Done (DoD)

Answer:

The **Definition of Done (DoD)** is a shared agreement that defines the quality standards a Product Backlog item must satisfy before it is considered complete. It ensures that every Increment is fully developed, tested, documented, and meets the required quality criteria. ***The Definition of Done promotes consistency, transparency, and high-quality product delivery throughout the project.***

🛡️ Chapter - Security

📝 Chapter - Documentation

🥂 Final